

Healthcare Claims Analytics

SQL + Tableau Business Intelligence Solution

Healthcare organizations process thousands of insurance claims every day, generating large volumes of financial and operational data. Without centralized reporting, identifying reimbursement patterns, monitoring provider activity, and understanding claim behavior becomes a complex and time-consuming task.

This project presents an end-to-end healthcare analytics solution that integrates SQL-based data transformation in Google BigQuery with interactive business intelligence visualizations in Tableau. Raw inpatient and outpatient Medicare claims were consolidated into a unified analytical dataset, enabling the creation of executive-level KPIs and interactive dashboards to support operational monitoring, executive reporting, and data-driven decision-making.

The final solution provides stakeholders with a comprehensive view of reimbursement spending, provider activity, claim distribution, and hospitalization metrics through an intuitive dashboard designed for business users.

1. BUSINESS PROBLEM

- Limited visibility into reimbursement trends.
- Difficulty identifying providers associated with high reimbursement amounts.
- Lack of consolidated reporting across inpatient and outpatient services.

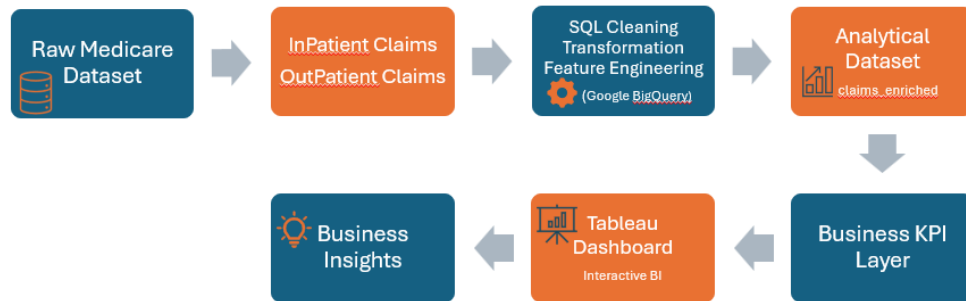
2. PROJECT OBJECTIVES

- Integrate inpatient and outpatient Medicare claims into a unified analytical dataset.
- Perform data cleaning and transformation using SQL in Google BigQuery.
- Design an interactive Tableau dashboard for executive reporting.
- Generate actionable business insights that support financial monitoring, operational performance evaluation, and provide a foundation for future fraud detection initiatives.

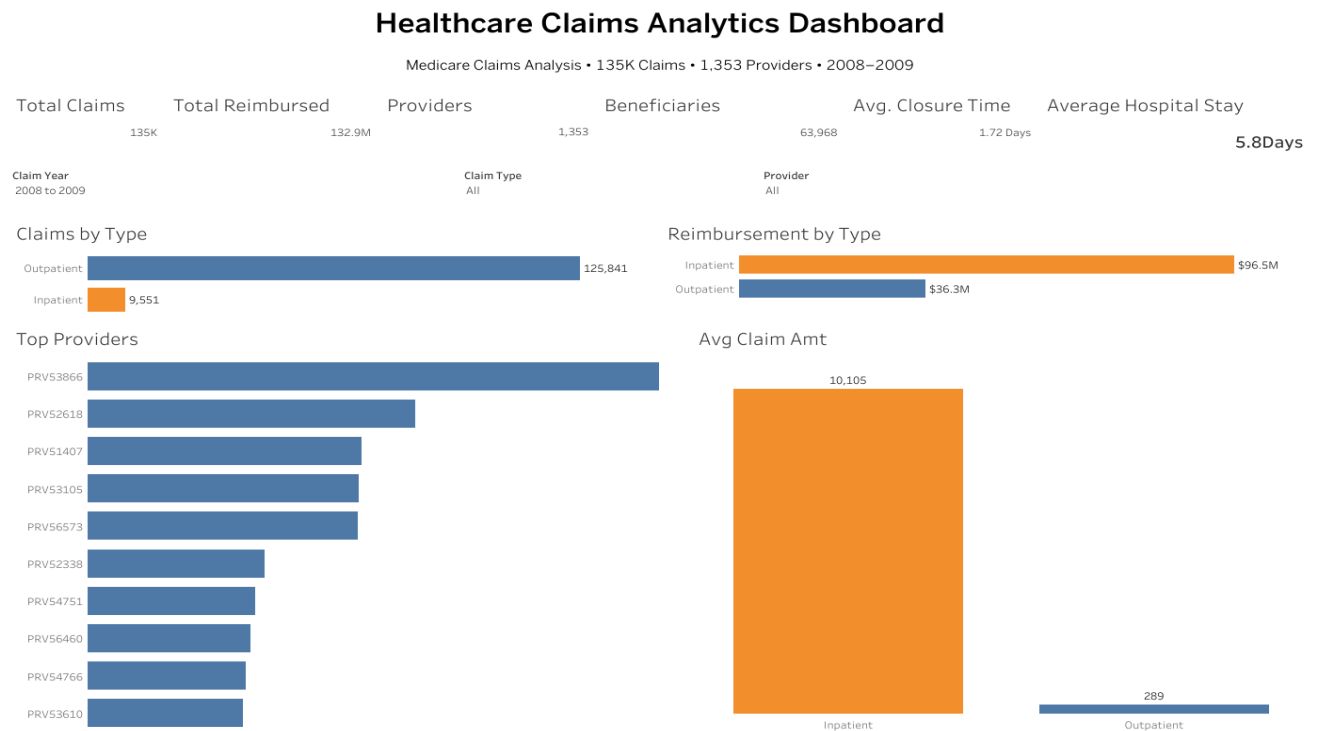
3. TECHNOLOGY STACK

Data	Processing	Visualization
BigQuery	SQL	Tableau

4. DATA PIPELINE



5. DASHBOARD



6. KEY PERFORMANCE INDICATORS

KPI	Value
Total Claims	135K
Total Reimbursed	\$132.9M
Providers	1,353
Beneficiaries	63,968
Average Hospital Stay	5.8 Days

7. KEY BUSINESS INSIGHTS

1. Inpatient claims account for the highest reimbursement spending.

Although inpatient claims represent only a small percentage of the total claims processed, they concentrate the majority of reimbursement expenditures. This finding highlights inpatient services as the primary driver of healthcare spending within the analyzed dataset.

2. Average reimbursement differs substantially by claim type.

The average reimbursement for inpatient claims exceeds \$10,000, while outpatient claims average approximately \$289. This significant difference emphasizes the importance of analyzing claim types separately when evaluating healthcare costs.

3. Reimbursement spending is concentrated among a limited number of providers.

The analysis reveals that a relatively small group of healthcare providers receives a disproportionately high share of total reimbursements. Monitoring these providers separately may improve financial oversight and support targeted auditing strategies.

4. Hospital stay duration establishes an operational benchmark.

The average inpatient stay is approximately 5.8 days, providing a useful baseline for operational performance monitoring. Future analyses may compare individual providers or patient groups against this benchmark to identify unusual hospitalization patterns.

8. SKILLS DEMONSTRATED

- SQL Data Transformation
- Google BigQuery
- Tableau Dashboard Development
- Business Intelligence
- KPI Design
- Data Cleaning & Preparation
- Healthcare Analytics
- Executive Reporting
- Business Storytelling
- Data Visualization Best Practices

Complete documentation, SQL scripts, Tableau workbook, and project repository are available through GitHub and Notion Portfolio.